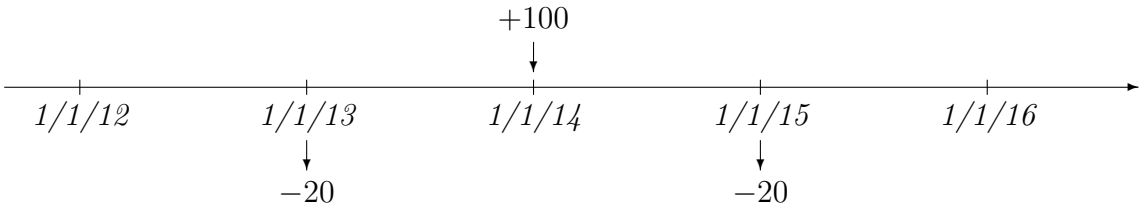
F78AA - Actuarial and Financial Mathematics A
Tutorial 8 – Measuring fund performance

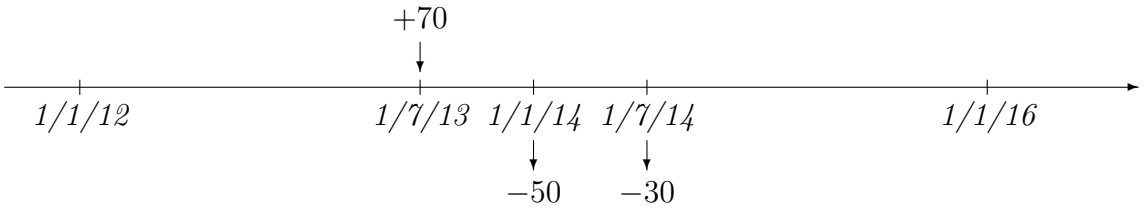
Question 1 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £170.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2014 was £100 just before the cash inflow on 1 Jan 2014 was received, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the two sub-periods calendar years 2012–2013 and 2014–2015.
3. If, in addition, the fund value was £110 on 1 January 2013 and £180 on 1 January 2015 (just before the cashflows on 1 January 2013 and 2015), calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 2 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £140.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2013 was £105, and on 1 January 2015 was £130, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the three sub-periods 2012, 2013–2014 and 2015.
3. Suppose, in addition to the information in part 2, that you knew the following fund values:

Date	1 July 2013	1 January 2014	1 July 2014
Fund Value	£115	£195	£155

Calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 3 [CT1, Apr 2010, Q10] A pension fund's assets were invested with two fund managers. On 1 January 2007, Manager A was given £120 000 and Manager B was given £100 000. A further £10 000 was invested with each manager on 1 January 2008 and again on 1 January 2009.

The values of the funds were:

Date	31 Dec 2007	31 Dec 2008	31 Dec 2009
Manager A	£130 000	£135 000	£180 000
Manager B	£140 000	£145 000	£150 000

1. Calculate the time-weighted rates of return earned by Manager A and Manager B over the period 1 January 2007 to 31 December 2009.

2. Show that the money-weighted rate of return earned by Manager A over the period 1 January 2007 to 31 December 2009 is approximately 9.4% per annum.

3. Explain, without performing further calculations, whether the money-weighted rate of return earned by Manager B over the period 1 January 2007 to 31 December 2009 was higher than, lower than or equal to that earned by Manager A.

4. Discuss the relative performance of the two fund managers.

Question 4 [CT1, Sep 2011, Q4] A pension fund makes the following investments in £m:

Date	1 Jan 2009	1 Jul 2009	1 Jan 2010
Investment amounts (£m)	1.5	6.0	4.0

The rates of return earned on money invested in the fund were as follows:

Time period	Rate of return (i.e. not annualised)
1 Jan 2009 to 30 Jun 2009	1%
1 Jul 2009 to 31 Dec 2009	2%
1 Jan 2010 to 31 Dec 2010	5%

Assume that 1 January to 30 June and 1 July to 31 December are precise half-year periods.

1. Calculate the time-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

2. Calculate the money-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

Solutions: Q1.1. 1.878%; Q1.2. 3.706%; Q1.3. 3.975%; Q2.1. 9.969%; Q2.2. 9.652%; Q2.3. 9.758%; Q3.1 A: 9.050% p.a. B: 9.409% p.a.; Q3.3. Lower; Q4.1. 4.005% p.a.; Q4.2 4.63% p.a.